LAB ASSIGNMENT – 9 (Mix Concepts)

DATE – 06.04.2026

NAME : Shreya Ghoshal
REGISTRATION NO. : AU/2023/0009045

ROLL NO. : UG/02/BTCSECSF/2023/003

SUBJECT : Unstructured Database Lab
SEC: E

COLLECTION 1: **members**

{

member_id: Number,

name: String,

plan: String,

city: String,

fee: Number

}

Table

member_id	name	plan	city	fee
1	Aarav	Gold	Delhi	5000
2	Meera	Silver	Mumbai	3000
3	Rohan	Gold	Bangalore	5500
4	Sneha	Platinum	Delhi	7000
5	Kunal	Silver	Chennai	3200

COLLECTION 2: **sessions**

{

session_id: Number,

member_id: Number,

```
  trainer: String,
  duration: Number,
  calories: Number
}
```

Table

session_id	member_id	trainer	duration	calories
101	1	Raj	60	400
102	2	Simran	45	300
103	3	Raj	70	500
104	4	Aman	80	600
105	5	Raj	50	350

```
test> use gymDB
switched to db gymDB
gymDB> db.createCollection("members")
... db.createCollection("sessions")
{ ok: 1 }
gymDB> db.members.insertMany([
...   {member_id: 1, name: "Aarav", plan: "Gold", city: "Delhi", fee: 5000},
...   {member_id: 2, name: "Meera", plan: "Silver", city: "Mumbai", fee: 3000},
...   {member_id: 3, name: "Rohan", plan: "Gold", city: "Bangalore", fee: 5500},
...   {member_id: 4, name: "Sneha", plan: "Platinum", city: "Delhi", fee: 7000},
...   {member_id: 5, name: "Kunal", plan: "Silver", city: "Chennai", fee: 3200}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d350be81c41e5d071e2621'),
    '1': ObjectId('69d350be81c41e5d071e2622'),
    '2': ObjectId('69d350be81c41e5d071e2623'),
    '3': ObjectId('69d350be81c41e5d071e2624'),
    '4': ObjectId('69d350be81c41e5d071e2625')
  }
}
gymDB> db.sessions.insertMany([
...   {session_id: 101, member_id: 1, trainer: "Raj", duration: 60,calories: 400},
...   {session_id: 102, member_id: 2, trainer: "Simran", duration: 45, calorcalories: 300},
...   {session_id: 103, member_id: 3, trainer: "Raj", duration: 70,calories: 500},
...   {session_id: 104, member_id: 4, trainer: "Aman", duration: 80, calorieccalories: 600},
...   {session_id: 105, member_id: 5, trainer: "Raj", duration: 50,calories: 350}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69d350c581c41e5d071e2626'),
    '1': ObjectId('69d350c581c41e5d071e2627'),
    '2': ObjectId('69d350c581c41e5d071e2628'),
    '3': ObjectId('69d350c581c41e5d071e2629'),
    '4': ObjectId('69d350c581c41e5d071e262a')
  }
}
```

QUESTIONS

Q1. Display all members

```
gymDB> db.members.find()
[
  {
    _id: ObjectId('69d350be81c41e5d071e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2622'),
    member_id: 2,
    name: 'Meera',
    plan: 'Silver',
    city: 'Mumbai',
    fee: 3000
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 5500
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2625'),
    member_id: 5,
    name: 'Kunal',
    plan: 'Silver',
    city: 'Chennai',
    fee: 3200
  }
]
```

Q2. Show name and plan only.

```
gymDB> db.members.find({}, {name: 1, plan: 1, _id: 0})
[
  { name: 'Aarav', plan: 'Gold' },
  { name: 'Meera', plan: 'Silver' },
  { name: 'Rohan', plan: 'Gold' },
  { name: 'Sneha', plan: 'Platinum' },
  { name: 'Kunal', plan: 'Silver' }
]
```

Q3. Find members from Delhi.

```
gymDB> db.members.find({city: "Delhi"})
[
  {
    _id: ObjectId('69d350be81c41e5d071e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  }
]
```

Q4. Find sessions with duration > 60.

```
gymDB> db.sessions.find({duration: {$gt: 60}})
[
  {
    _id: ObjectId('69d350c581c41e5d071e2628'),
    session_id: 103,
    member_id: 3,
    trainer: 'Raj',
    duration: 70,
    calories: 500
  },
  {
    _id: ObjectId('69d350c581c41e5d071e2629'),
    session_id: 104,
    member_id: 4,
    trainer: 'Aman',
    duration: 80,
    calories: 600
  }
]
```

Q5. Sort members by fee descending.

```
gymDB> db.members.find().sort({fee: -1})
[
  {
    _id: ObjectId('69d350be81c41e5d071e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 5500
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5000
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2625'),
    member_id: 5,
    name: 'Kunal',
    plan: 'Silver',
    city: 'Chennai',
    fee: 3200
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2622'),
    member_id: 2,
    name: 'Meera',
    plan: 'Silver',
    city: 'Mumbai',
    fee: 3000
  }
]
```

Q6. Update fee +500 for Gold members.

```
gymDB> db.members.updateMany(
...   {plan: "Gold"},
...   {$inc: {fee: 500}}
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
```

Q7. Delete members with fee < 3500.

```
gymDB> db.members.deleteMany({fee: {$lt: 3500}})
{ acknowledged: true, deletedCount: 2 }
gymDB> db.members.insertOne({
...   member_id: 6,
...   name: "Priya",
...   plan: "Gold",
...   city: "Kolkata",
...   fee: 4000
... })
{
  acknowledged: true,
  insertedId: ObjectId('69d3511781c41e5d071e262b')
}
```

Q8. Insert a new member.

```
gymDB> db.members.insertOne({
...   member_id: 6,
...   name: "Priya",
...   plan: "Gold",
...   city: "Kolkata",
...   fee: 4000
... })
{
  acknowledged: true,
  insertedId: ObjectId('69d3539181c41e5d071e262c')
}
```

Q9. Average fee per plan.

```
gymDB> db.members.aggregate([
...   {$group: {_id: "$plan", avgFee: {$avg: "$fee"}}}
... ])
[
  { _id: 'Gold', avgFee: 5166.666666666667 },
  { _id: 'Platinum', avgFee: 7000 }
]
```

Q10. Total calories burned per trainer.

```
gymDB> db.sessions.aggregate([
...   {$group: {_id: "$trainer", totalCalories: {$sum: "$calories"}}}
... ])
[
  { _id: 'Simran', totalCalories: 300 },
  { _id: 'Raj', totalCalories: 1250 },
  { _id: 'Aman', totalCalories: 600 }
]
```

Q11. Maximum fee per city.

```
gymDB> db.members.aggregate([
...   {$group: {_id: "$city", maxFee: {$max: "$fee"}}}
... ])
[
  { _id: 'Kolkata', maxFee: 4000 },
  { _id: 'Delhi', maxFee: 7000 },
  { _id: 'Bangalore', maxFee: 6000 }
]
```

Q12. Count members per plan.

```
gymDB> db.members.aggregate([
...   {$group: {_id: "$plan", count: {$sum: 1}}}
... ])
[ { _id: 'Gold', count: 3 }, { _id: 'Platinum', count: 1 } ]
```

Q13. Trainers with avg calories > 400.

```
gymDB> db.sessions.aggregate([
...   {$group: {_id: "$trainer", avgCalories: {$avg: "$calories"}}},
...   {$match: {avgCalories: {$gt: 400}}}
... ])
[
  { _id: 'Aman', avgCalories: 600 },
  { _id: 'Raj', avgCalories: 416.6666666666667 }
]
```

Q14. Total duration per member_id.

```
gymDB> db.sessions.aggregate([
...   {$group: {_id: "$member_id", totalDuration: {$sum: "$duration"}}}
... ])
[
  { _id: 4, totalDuration: 80 },
  { _id: 3, totalDuration: 70 },
  { _id: 1, totalDuration: 60 },
  { _id: 2, totalDuration: 45 },
  { _id: 5, totalDuration: 50 }
]
```

Q15. Join members with sessions.

```
gymDB> db.members.aggregate([
...   {
...     $lookup: {
...       from: "sessions",
...       localField: "member_id",
...       foreignField: "member_id",
...       as: "sessions"
...     }
...   }
... ])
[
  {
    _id: ObjectId('69d350be81c41e5d071e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500,
    sessions: [
      {
        _id: ObjectId('69d350c581c41e5d071e2626'),
        session_id: 101,
        member_id: 1,
        trainer: 'Raj',
        duration: 60,
        calories: 400
      }
    ]
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000,
    sessions: [
      {
        _id: ObjectId('69d350c581c41e5d071e2628'),
        session_id: 103,
        member_id: 3,
        trainer: 'Raj',
        duration: 70,
        calories: 500
      }
    ]
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000,
```

```
    sessions: [
      {
        _id: ObjectId('69d350c581c41e5d071e2629'),
        session_id: 104,
        member_id: 4,
        trainer: 'Aman',
        duration: 80,
        calories: 600
      }
    ]
  },
  {
    _id: ObjectId('69d3511781c41e5d071e262b'),
    member_id: 6,
    name: 'Priya',
    plan: 'Gold',
    city: 'Kolkata',
    fee: 4000,
    sessions: []
  }
]
```

Q16. Show members who have sessions.

```
gymDB> db.members.aggregate([
...   {
...     $lookup: {
...       from: "sessions",
...       localField: "member_id",
...       foreignField: "member_id",
...       as: "sessions"
...     }
...   },
...   {
...     $match: {sessions: {$ne: []}}
...   }
... ])
[
  {
    _id: ObjectId('69d350be81c41e5d071e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500,
    sessions: [
      {
        _id: ObjectId('69d350c581c41e5d071e2626'),
        session_id: 101,
        member_id: 1,
        trainer: 'Raj',
        duration: 60,
        calories: 400
      }
    ]
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000,
    sessions: [
      {
        _id: ObjectId('69d350c581c41e5d071e2628'),
        session_id: 103,
        member_id: 3,
        trainer: 'Raj',
        duration: 70,
        calories: 500
      }
    ]
  },
  {
```

```
  {
    _id: ObjectId('69d350be81c41e5d071e2624'),
    member_id: 4,
    name: 'Sneha',
    plan: 'Platinum',
    city: 'Delhi',
    fee: 7000,
    sessions: [
      {
        _id: ObjectId('69d350c581c41e5d071e2629'),
        session_id: 104,
        member_id: 4,
        trainer: 'Aman',
        duration: 80,
        calories: 600
      }
    ]
  }
]
```

Q17. Create index on city.

```
gymDB> db.members.createIndex({city: 1})
city_1
```

Q19. Create compound index on plan and fee.

```
gymDB> db.members.createIndex({plan: 1, fee: -1})
plan_1_fee_-1
```

Q20. Display 2nd and 3rd highest fee members.

```
gymDB> db.members.find()
...    .sort({fee: -1})
...    .skip(1)
...    .limit(2)
[
  {
    _id: ObjectId('69d350be81c41e5d071e2623'),
    member_id: 3,
    name: 'Rohan',
    plan: 'Gold',
    city: 'Bangalore',
    fee: 6000
  },
  {
    _id: ObjectId('69d350be81c41e5d071e2621'),
    member_id: 1,
    name: 'Aarav',
    plan: 'Gold',
    city: 'Delhi',
    fee: 5500
  }
]
```